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# QMS-FLOW
# SYSTEM DOCUMENTATION

Presentation Video link: https://youtu.be/OvvQcK3gM4s

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# Test Case

Test cases are structured, step-by-step validations designed to verify that a specific system function behaves as defined in the requirements. They translate design artifacts such as sequence diagramsinto executable checks, ensuring that both the intended workflows and alternate or exception paths are systematically examined. By defining preconditions, test data, actions, and expected outcomes, test cases provide a repeatable and measurable way to confirm software quality, reduce defects, and align development with stakeholder expectations.

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# TC001: TEST <AUDIT MANAGEMENT> SUBSYSTEM

# TC001_01: TEST <SCENARIO OF SCHEDULE AUDIT (SD001)>

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# TC001_01_01: Test <Normal scenario of Schedule Audit (SD001)>

| | | | | | |
|---|---|---|---|---|---|
| H | TC001_01_01 | Test Case Description | Verify that Quality Manager can shcedule a new audit with valid inputs and assign an available auditor | | |
| Created By | MUHAMMAD LUQMAN BIN MOHD AZMI | Reviewed By | | Version | 1.0 |
| QA Tester's Log | | | | | |
| Tester's Name | | Date Tested | | Test Case (Pass/Fail/Not) | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager is logged in with valid credentials |
| 2 | Auditor exists in the system |
| 3 | Audit Management module is accessible |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Audit Title: "Internal Audit Production Q1 2025" |
| 2 | Department: "Production" |
| 3 | Scheduled Date : "15/03/2025" |
| 4 | Auditor: "John Doe" |

**Test Scenario** User navigates to the Audit Management module, fills in all required audit details with valid future date and available auditor, and successfully schedules the audit.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Open Schedule Audit form | Form loads with required fields | | |
| 2 | Fill in audit details and select auditor | Data entered, auditor is shown | | |
| 3 | Click "Save Audit" | Display "Audit scheduled succesfully" message | | |
| 4 | Check audit in calendar | Audit appears with status "Scheduled" | | |

# TC001_01_02: Test <Alternate Scenario - Auditor unavailable (SD001)>

| | | | | | |
|---|---|---|---|---|---|
| Test Case ID | TC001_01_02 | Test Case Description | Verify system prevents scheduling if auditor is already booked. | | |
| Created By | MUHAMMAD LUQMAN BIN MOHD AZMI | Reviewed By | | Version | 1.0 |
| QA Tester's Log | | | | | |
| Tester's Name | | Date Tested | | Test Case (Pass/Fail/Not) | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager logged in. |
| 2 | Auditor "John Doe" already has audit on 15/03/2025. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Title: Q1 Production Audit |
| 2 | Department: Production |
| 3 | Date: 15/03/2025 |
| 4 | Auditor: John Doe |

| | |
|---|---|
| Test Scenario | User attempts to schedule an audit with an auditor who is already booked on the same date; system displays an error and prevents submission. |

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Open Schedule Audit form. | Form loads. | | |
| 2 | Enter details, select auditor (already booked). | Auditor selected. | | |
| 3 | Click "Save Audit". | Error: "Selected auditor is unavailable." | | |
| 4 | Form remains open. | User can pick another auditor. | | |

# TC001_01_03: Test <Alternate Scenario - Past Date Validation (SD001)>

| | | | | | |
|---|---|---|---|---|---|
| Test Case ID | TC001_01_03 | Test Case Description | Verify system rejects past audit dates. | | |
| Created By | MUHAMMAD LUQMAN BIN MOHD AZMI | Reviewed By | | Version | |
| QA Tester's Log | | | | | |
| Tester's Name | | Date Tested | | Test Case (Pass/Fail/Not) | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager logged in. |
| 2 | |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Title: Past Audit Test |
| 2 | Department: Production |
| 3 | Date: 01/01/2024 |
| 4 | Auditor: Raymond Lim |

**Test Scenario** User enters a past date while scheduling an audit; system rejects the input with an appropriate error message and highlights the date field.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Open Schedule Audit form. | Form loads. | | |
| 2 | Enter past date in date field. | Date entered. | | |
| 3 | Click "Save Audit". | Error: "Audit date cannot be in the past." | | |
| 4 | Form stays open. | Date field highlighted. | | |

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# TC002: TEST <CHANGE CONTROL> SUBSYSTEM

# TC002_01: TEST <MANAGE DOCUMENTS (SD003)>

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# TC002_02_01: Test<Test Normal Upload Scenario (SD004)>

| | | | | | |
|---|---|---|---|---|---|
| | TC002_01_01 | **Test Case Description** | Verify successful upload of a valid document file. | | |
| **Created By** | SEAH ZHANG JIAN | **Reviewed By** | | **Version** | |

**QA Tester's Log**

| | | | | | |
|---|---|---|---|---|---|
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager is logged in successfully. |
| 2 | User is on the "Upload Document" screen. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Title: "Q1 Safety SOP" |
| 2 | Dept: "Production" |
| 3 | File: "SOP_v1.pdf" (5MB) |
| 4 | |

| | |
|---|---|
| **Test Scenario** | Verify successful creation of a "Draft" document when a valid file and all metadata are provided. |

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Enter a valid "Document Title". | System displays a success message: "Document uploaded successfully as Draft." | | |
| 2 | Select a valid "Department" from the dropdown. | The document appears in the "My Documents" list with status "DRAFT". | | |
| 3 | Click "Browse" and select a valid PDF file (e.g., under 10MB). | Database record is created in Document table. | | |
| 4 | Click the "Submit Upload" button. | The system displays success message to the user | | |

# TC002_02_02: Test<Test Alternate Scenario–Invalid File Format Selected (SD004)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC002_01_02 | **Test Case Description** | Verify validation error when uploading an invalid file format. | | |
| **Created By** | SEAH ZHANG JIAN | **Reviewed By** | | **Version** | |

**QA Tester's Log**

| | | | | | |
|---|---|---|---|---|---|
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager is logged in. |
| 2 | User is on the upload screen. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | File: "setup.exe" |
| 2 | |
| 3 | |
| 4 | |

| | |
|---|---|
| **Test Scenario** | Verify the system rejects uploads of unsupported or unsafe file types (e.g., .exe). |

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Enter valid metadata. | System displays an error message: "Invalid file format. Only PDF and DOCX allowed." | | |
| 2 | Click "Browse" and select an invalid file type (e.g., .exe, .jpg). | The file is not uploaded to the server. | | |
| 3 | Click "Submit Upload". | The system displays success message to the user | | |

# TC002_02_03: Test<Test Alternate Scenario–Missing Mandatory Metadata (SD004)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC002_01_03 | **Test Case Description** | Verify validation error when mandatory metadata is missing. | | |
| **Created By** | SEAH ZHANG JIAN | **Reviewed By** | | **Version** | |

**QA Tester's Log**

| | | | | | |
|---|---|---|---|---|---|
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager is logged in. |
| 2 | User is on the upload screen. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Title: "" |
| 2 | File: "SOP_v1.pdf" |
| 3 | |
| 4 | |

| | |
|---|---|
| **Test Scenario** | Verify the form submission is blocked if required fields like Title are empty. |

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Leave the "Document Title" field blank. | System displays a validation error under the Title field: "Document Title is required." | | |
| 2 | Select a valid file. | The submission is blocked. | | |
| 3 | Click "Submit Upload". | The system displays success message to the user | | |

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# TC003: TEST <CUSTOMER COMPLAINTS> SUBSYSTEM

# TC003_02: TEST <SCENARIO OF INVESTIGATE COMPLAINT (SD006)>

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# TC003_02_01: Test <Normal scenario of Investigate Complaint (SD006)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC003_02_01 | **Test Case Description** | Verify successful assignment of an investigator to a pending complaint. | | |
| **Created By** | Tan Hui Shan | **Reviewed By** | | **Version** | |
| **QA Tester's Log** | | | | | |
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | A complaint with "Pending" status exists. |
| 2 | User is on the ComplaintPage. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | complaintID: "CMP-001" |
| 2 | investigator: "John Doe" |
| 3 | targetDate: "2026-05-20" (Future date) |
| 4 | |

**Test Scenario** Verify that the system updates the record and triggers a notification when a valid investigator and future target date are assigned to a pending complaint.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Select complaint "CMP-001" from the pending list. | ComplaintRecord status updates | | |
| 2 | Enter "John Doe" and target date, then click confirm. | NotificationSystem alerts investigator | | |
| 3 | | UI displays "Assignment confirmed" | | |
| 4 | | | | |

# TC003_02_02: Test <Alternate scenario - Invalid Date (SD006)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC003_02_02 | **Test Case Description** | Verify that the system rejects assignment dates that are in the past. | | |
| **Created By** | Tan Hui Shan | **Reviewed By** | | **Version** | |
| **QA Tester's Log** | | | | | |
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | User is on the investigation assignment screen |
| 2 | |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | targetDate: "2020-01-01" (Past date) |
| 2 | |
| 3 | |
| 4 | |

| | |
|---|---|
| **Test Scenario** | Verify that the assignment is blocked and an error message appears if a past date is entered in the target date field. |

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Input a past date into the targetDate field and attempt to confirm. | ComplaintController triggers the [Invalid Date] path | | |
| 2 | | UI displays "Select a future date". | | |
| 3 | | | | |
| 4 | | | | |

# TC003_02_03: Test <Alternate scenario - System Error (SD006)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC003_02_03 | **Test Case Description** | Verify system behavior during a database timeout during assignment update | | |
| **Created By** | Tan Hui Shan | **Reviewed By** | | **Version** | |

**QA Tester's Log**

| | | | | | |
|---|---|---|---|---|---|
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | Simulate a network timeout/latency issue |
| 2 | |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Valid investigator assignment data |
| 2 | |
| 3 | |
| 4 | |

**Test Scenario** Verify that the system displays a "System unavailable" message and prevents record corruption during a database connection timeout.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Click "Confirm Assignment" | ComplaintDataAccess returns "Database connection timed out" | | |
| 2 | | UI triggers [System Error] path and shows "System unavailable, try again" | | |
| 3 | | | | |
| 4 | | | | |

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# TC004: TEST <CAPA MANAGEMENT> SUBSYSTEM

# TC004_01: TEST <SCENARIO OF INITIATE CORRECTIVE ACTION (SD007)>

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# TC004_01_01: Test <Normal scenario of Initiate CAPA (SD007)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC004_01_01 | **Test Case Description** | Validate that a Quality Manager can successfully initiate a new CAPA record with valid data. | | |
| **Created By** | Lee Hui Zhen | **Reviewed By** | | **Version** | 1.0 |
| **QA Tester's Log** | | | | | |
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | User is logged in as "Quality Manager". |
| 2 | A valid Audit Finding exists. |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Source ID: AUD-001 |
| 2 | Root Cause: “Machine calibration error detected.” |
| 3 | Owner: “John Doe” |
| 4 | |

**Test Scenario** User navigates to the CAPA module, selects an existing audit finding, fills in all mandatory fields with valid data, and submits the form successfully.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Click “Initiate New CAPA” on the dashboard. | System displays the “Initiate CAPA” form. | | |
| 2 | Enter Source ID, Root Cause, Owner. | System accepts input and validates character limits. | | |
| 3 | Click “Submit”. | CAPA record saved; Source status updated to “CAPA Initiated”; success message displayed. | | |
| 4 | Search for new CAPA ID (e.g., CAPA-001). | Record visible with status “Open”. | | |

# TC004_01_02: Test <Alternate scenario – Invalid Source ID (SD007)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC004_01_02 | **Test Case Description** | Verify that the system prevents CAPA initiation when the Quality Manager enters an invalid Source ID. | | |
| **Created By** | Lee Hui Zhen | **Reviewed By** | | **Version** | 1.0 |

**QA Tester's Log**

| | | | | | |
|---|---|---|---|---|---|
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | Quality Manager is authenticated and logged in. |
| 2 | The Source ID entered does not exist in the system (e.g., no matching Audit Finding or Complaint record). |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Source ID: AUD-999 (invalid) |
| 2 | Root Cause: “Calibration error” |
| 3 | Action Plan: “Recalibrate equipment and retrain staff” |
| 4 | Deadline: 2026-02-01 |
| 5 | Owner: “John Doe” |

**Test Scenario** User attempts to initiate a CAPA with an invalid Source ID.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Navigate to CAPA module | CAPA initiation form loads | | |
| 2 | Enter invalid Source ID (AUD-999) along with Root Cause, Action Plan, Deadline, Owner | System displays error “Invalid Source ID” and prevents submission. | | |
| 3 | Click “Submit” | Error message displayed: “Invalid Source ID”; CAPA record not saved | | |

# TC004_01_03: Test <Alternate scenario – Missing Root Cause (SD007)>

| | | | | | |
|---|---|---|---|---|---|
| **Test Case ID** | TC004_01_03 | **Test Case Description** | Verify system behavior when mandatory fields (Root Cause) are left empty. | | |
| **Created By** | Lee Hui Zhen | **Reviewed By** | | **Version** | 1.0 |
| **QA Tester's Log** | | | | | |
| **Tester's Name** | | **Date Tested** | | **Test Case (Pass/Fail/Not)** | |

| S # | Prerequisites: |
|---|---|
| 1 | User is logged in as Quality Manager. |
| 2 | |
| 3 | |
| 4 | |

| S # | Test Data |
|---|---|
| 1 | Source ID: AUD-001 |
| 2 | Root Cause: [Empty] |
| 3 | Owner: “John Doe” |
| 4 | |

**Test Scenario** User attempts to submit the CAPA form without entering the mandatory Root Cause field to verify validation logic.

| Step # | Step Details | Expected Results | Actual Results | Pass / Fail / Not executed / Suspended |
|---|---|---|---|---|
| 1 | Navigate to Initiate Page. | System displays the “Initiate CAPA” form. | | |
| 2 | Enter Source ID and Owner, leave Root Cause empty. | System detects missing mandatory field. | | |
| 3 | Submit Form. | Record not saved; error message “Root Cause is required” displayed. | | |

# Thank you

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